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Distributive laws for relative monads

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This talk is based on work in progress joint with JS Lemay on distributive laws for relative monads. Relative monads form a generalization of monads where the underlying functor may have distinct domain and codomain categories. This makes it possible to study monad constructions beyond the classical setting of endofunctors and thereby provides a convenient language for algebraic structures. Given two ordinary monads, a distributive law between them is a natural transformation satisfying specific compatibility conditions with the monad structures which express abstractly that the two monads distribute one over the other. The data of a distributive law is equivalent to the data of an extension of one monad to the Kleisli category of the other one. Motivated by this latter characterization, I will present a notion of a distributive law between relative monads and discuss to which extent this recovers the non-relative case. Moreover, I will illustrate this concept of a relative distributive law by an example involving the Kleisli and Eilenberg-Moore categories of ordinary monads.

Keywords: Relative monads; distributive laws; Kleisli category
Subject area: Relative monads (category theory)